NAEEM NAIRO

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SUMMARY

Biomedical engineer (M.S., Fulbright Scholar) with 3+ years taking mobility and juvenile products from concept to mass production. Certified SOLIDWORKS Expert in CAD, FEA, DFM, GD&T, prototyping and verification testing, with hands-on EMG instrumentation and embedded systems. Seeking internship and full-time roles in medical device R&D, biomedical instrumentation and quality engineering.

EXPERIENCE

Product Design Engineer, AHL Logics *Nov 2021 – Aug 2025*

Full time from Jun 2022; part time Nov 2021 – Jun 2022

- Developed 4 products from concept through prototyping and testing; 3 reached mass production and launch.
- Co-inventor, as part of the design team, on 5 U.S. patents pending for a foldable multi-child stroller wagon; designed its folding mechanism for easy, aligned folding.
- Reduced structural part weight by 15%+ on an infant car seat and stroller using FEA, offsetting weight added by new features; validated predictions against physical prototype tests.
- Designed the attachment mechanism for a clip-on e-bike conversion drive; built and tested prototypes and applied DFM for production.
- Folding walker for elderly users (prototype): validated the mechanism, built the load map, ran FEA safety checks, compared magnesium and carbon fiber with aluminum with DFM redesigns, and coordinated ISO 11199 and internal testing.
- Wrote design specs, test plans and reports, drawings and BOMs; led design transfer with manufacturing, quality and suppliers; supported FMEA, root-cause analysis and engineering changes.

Teaching Assistant, Instrumentation Laboratory, University of New Haven *Fall 2025 – Present*

- Guide students through LabVIEW and NI ELVIS experiments in EMG signal acquisition and temperature sensing.
- Adapting a ball-and-beam control platform into a rehabilitation tool with training strategies for post-stroke mobility impairments.

PROJECTS

NeuroGrip: EMG-Controlled Robotic Hand *Python, PyTorch, Arduino, SOLIDWORKS*

- Built an 11-DOF 3D-printed robotic hand controlled in real time from 8-channel forearm EMG through a neural network trained in PyTorch.
- Stopped output flicker with a temporal-consistency loss term; built a kinematic simulator from the CAD assembly to check joint ranges.

EyeMotion: Eye-Gaze Proportional Control *Python, MediaPipe, OpenCV*

- Turned a standard webcam into a hands-free proportional controller using eye landmarks, evaluated on a ball-and-beam task; manuscript under review.

GrapGrow: Modular Pediatric Prosthetic Arm *SOLIDWORKS, 3D printing*

- Designed a parametric, 3D-printable arm with swappable segments sized from anthropometric data for ages 8–14, so it can grow with the child.

EDUCATION

M.S. Biomedical Engineering, University of New Haven *GPA 3.9 · Fulbright Scholar*

Coursework: Biomedical Sensors and Instrumentation, Signal Analysis, Biomechanics, Biostatistics, Biomedical Polymers

B.S. Mechanical Engineering (Mechatronics), Palestine Polytechnic University

PUBLICATION

M. N. Nairokh. Hands-Free Proportional Control Through Eye Gaze: A Low-Cost Webcam Interface Evaluated on a Ball-and-Beam Task. Under review, 2026.

SKILLS

Design and prototyping: SOLIDWORKS (CSWE), CAD, DFM, GD&T, mechanism design, 3D printing and rapid prototyping

Analysis and simulation: SOLIDWORKS Simulation, ANSYS, structural FEA, MATLAB, DOE, statistical analysis (Python, R)

Instrumentation and embedded: EMG acquisition and processing, LabVIEW, NI ELVIS, Arduino, ESP32, STM32, I2C, PyTorch

Quality and regulatory: design verification and validation, test plans and reports, FMEA, root-cause analysis, ISO 11199 testing; working knowledge of ISO 13485 and FDA 21 CFR Part 820

CERTIFICATIONS

SOLIDWORKS Design Expert (CSWE) · SOLIDWORKS Simulation Professional · SOLIDWORKS CAD Design Professional · SOLIDWORKS Additive Manufacturing Associate · Additive Manufacturing: Materials for 3D Printing

HONORS AND LEADERSHIP

Fulbright Scholar (2025–2027) · SOLIDWORKS Champion (2026) · Secretary, Biomedical Engineering Society, UNH (2026–) · President, Mechanical Engineering Society, PPU (2021) · Member: IEEE, ASME, BMES